

Ethanol Blending in Gasoline – Thailand

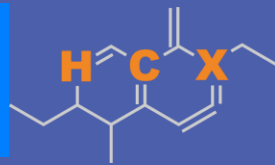
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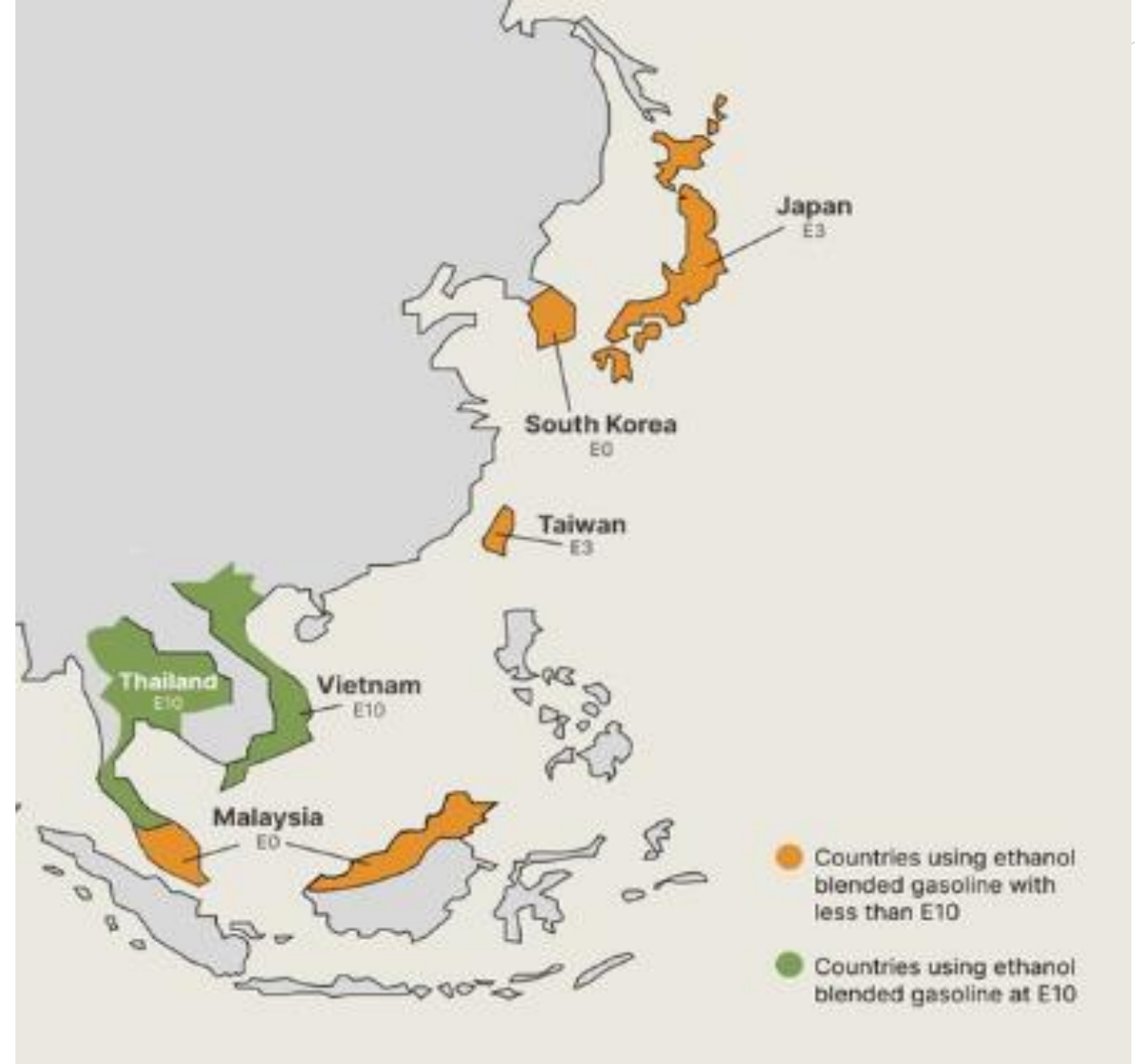


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Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Thailand



In 2024, gasoline consumption in Thailand was 11,452 million liters (3,025 million gallons) and growth rate was -0.5%. Gasoline production and net imports were 13,337 and -1,885 million liters (3,523 and -498 million gallons) correspondingly. Thailand offers two main grades: RON 91 (E10) and RON 95 (E10 and E20) with an average octane of 94.1 RON.

Thailand's bioethanol production primarily uses sugarcane and cassava as feedstocks, with approximately 30 facilities producing over 1 billion liters annually. The promotion of bio-based products like ethanol is part of Thailand's bioeconomy policy, supporting both the energy sector and domestic economic growth. In 2024, ethanol consumption supplied by local production in Thailand was 1,289 million liters (341 million gallons) representing 11.3% of gasoline demand.

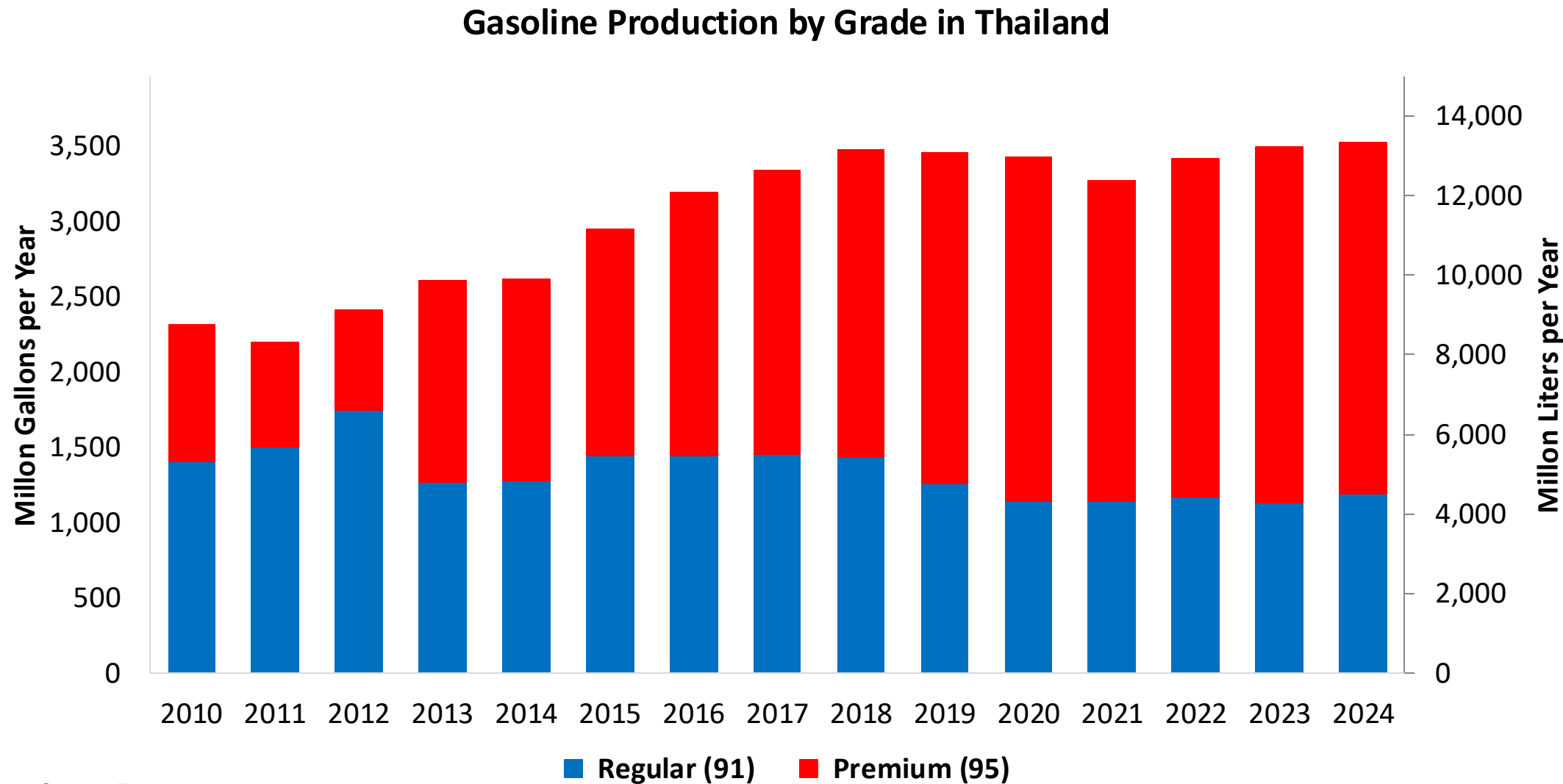
Source: Eppo, USDA, Department of Energy Business

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.

Source: Eppo

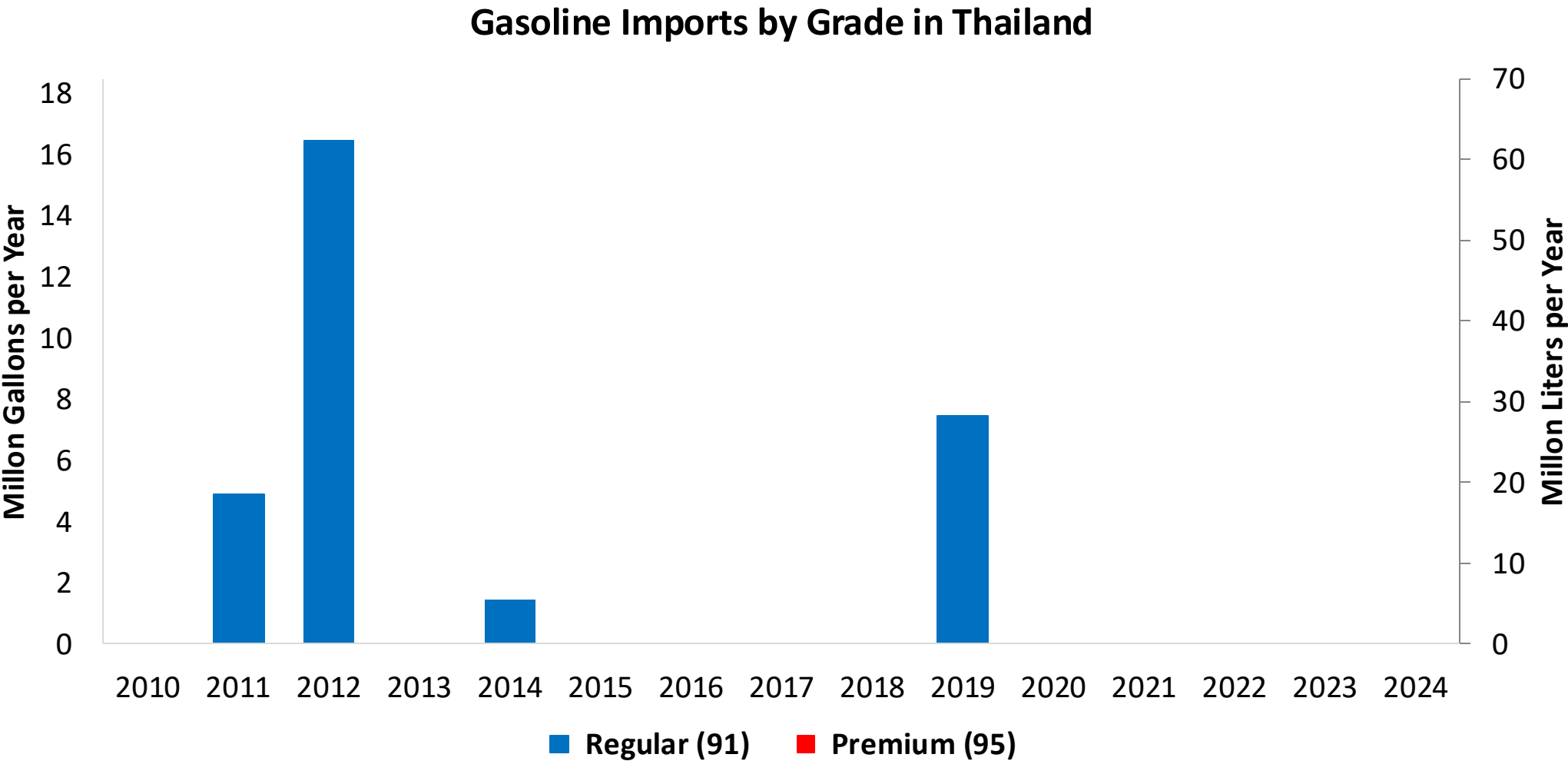
Year	Gasohol E10 (91) (Million Gallons)	Gasohol E10 (95) (Million Gallons)	Gasohol E20 (95) (Million Gallons)	Total (Million Gallons)
2010	1,180	560	50	1,790
2011	1,310	370	60	1,740
2012	1,400	300	80	1,780
2013	900	770	130	1,800
2014	940	770	140	1,850
2015	1,050	940	140	2,130
2016	1,070	1,130	150	2,350
2017	1,020	920	240	2,180
2018	950	1,120	240	2,310
2019	900	1,050	290	2,240
2020	780	1,450	210	2,440
2021	650	1,450	180	2,280
2022	670	1,400	220	2,290
2023	650	1,930	110	2,690
2024	700	1,930	110	2,740

Gasoline Production in Thailand



Source: Eppo

Gasoline Imports in Thailand



Source: Eppo

The FARO90 logo is shown in a blue box. To its right is a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.

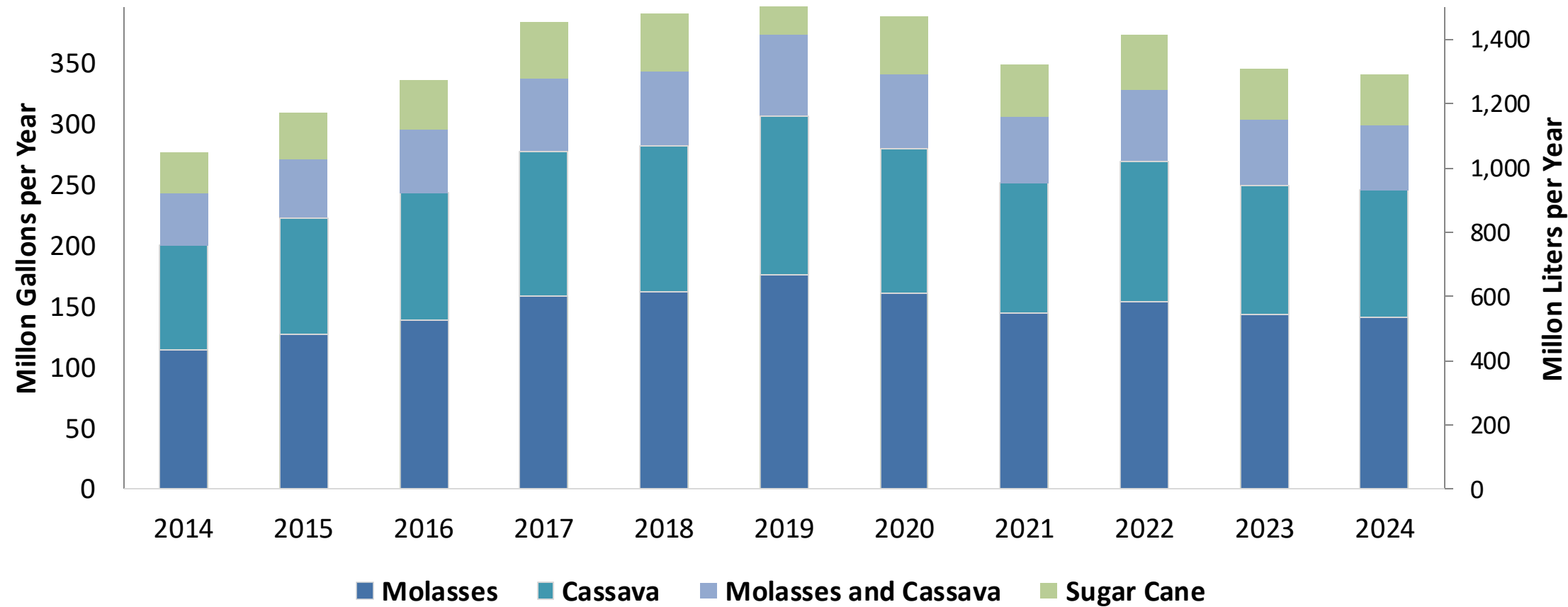
The chart displays the annual gasoline consumption in the United States from 2010 to 2024. The data is presented as stacked bars, where the blue portion represents Regular (91) gasoline and the red portion represents Premium (95) gasoline. The left Y-axis measures consumption in Million Gallons per Year (0 to 500), and the right Y-axis measures it in Million Liters per Year (0 to 1,800). The X-axis lists the years from 2010 to 2024. The total consumption shows a general upward trend, peaking in 2022 and 2024, with a notable dip in 2011 and 2019. Premium gasoline consumption remains a relatively small but consistent portion of the total.

Year	Regular (91) (Million Gallons)	Premium (95) (Million Gallons)	Total (Million Gallons)
2010	210	140	350
2011	205	65	270
2012	365	35	400
2013	355	85	440
2014	335	35	370
2015	380	25	405
2016	370	30	400
2017	425	15	440
2018	470	20	490
2019	350	15	365
2020	355	25	380
2021	475	5	480
2022	495	10	505
2023	465	15	480
2024	480	20	500

Source: Eno

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Ethanol Production and Demand in Thailand



Source: Eppo, USDA

Gasoline Quality in Thailand

Name	B.E. 2557 (2014)					
Implementation Date	2014					
Applicability	Nationwide					
Grade	Gasohol E10 - 91		Gasohol E10 - 95		Gasohol E20	
	Min	Max	Min	Max	Min	Max
RON	91.0		95.0		95.0	
MON	80.0		84.0		84.0	
AKI						
Benzene (%vol)		1.0		1.0		1.0
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)		0.005		0.005		0.005
Phosphorous (g/L)		0.013		0.013		0.013
Sulfur (ppm)		50		50		50
Oxygen (%w)						
Ethanol (%vol)		10.0		10.0		20.0
Other Oxygenates	The approval of the Director General of the Department of Energy Business					
PVR 37.8°C (kPa)		62		62		62

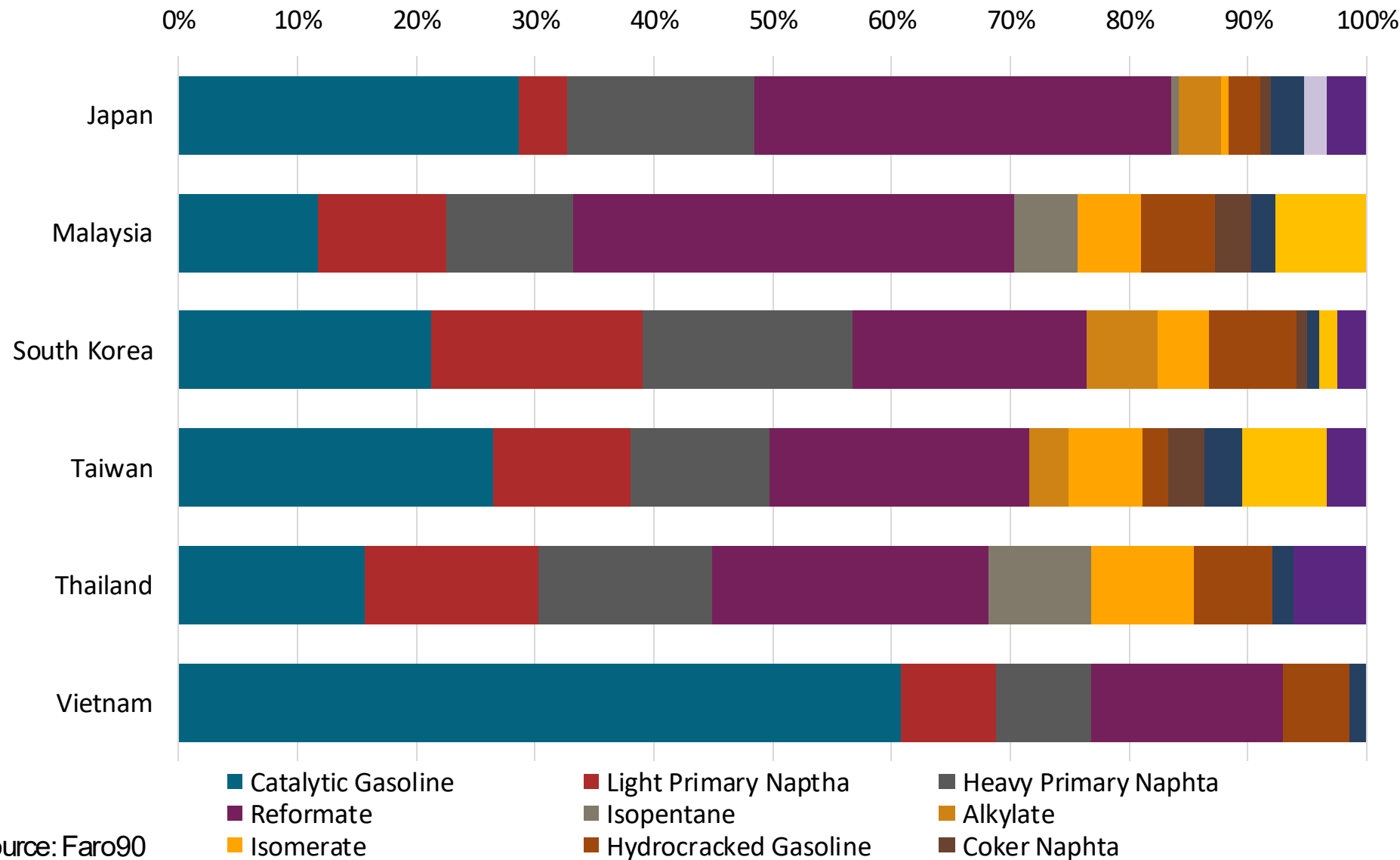
Source: Department of Energy Business

Gasoline Component Blending in Asia



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

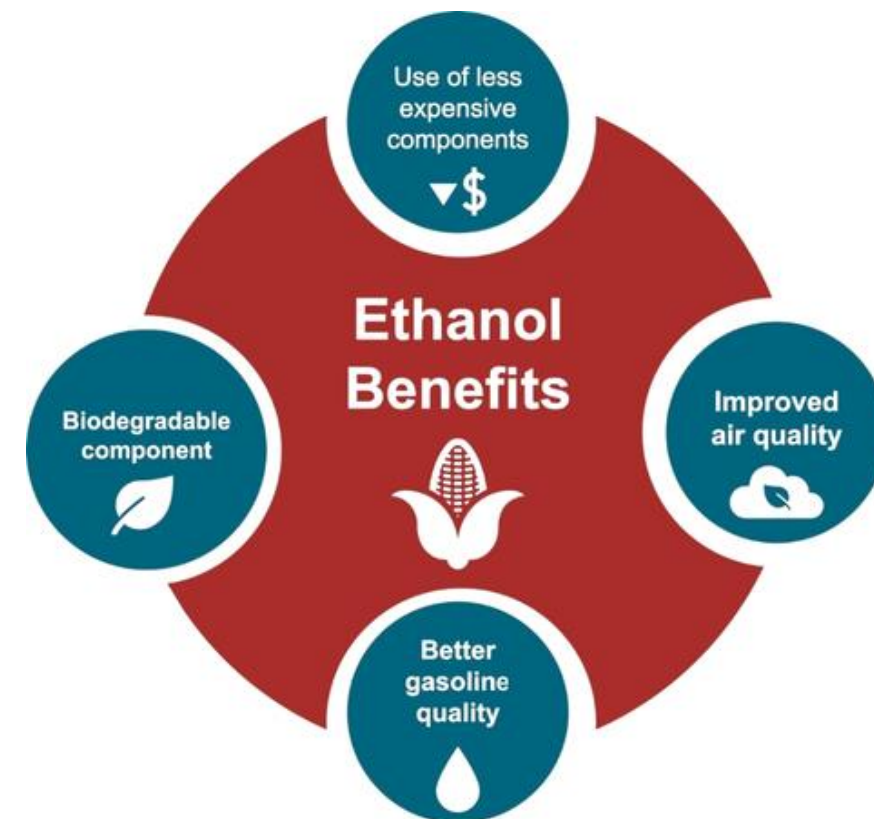
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

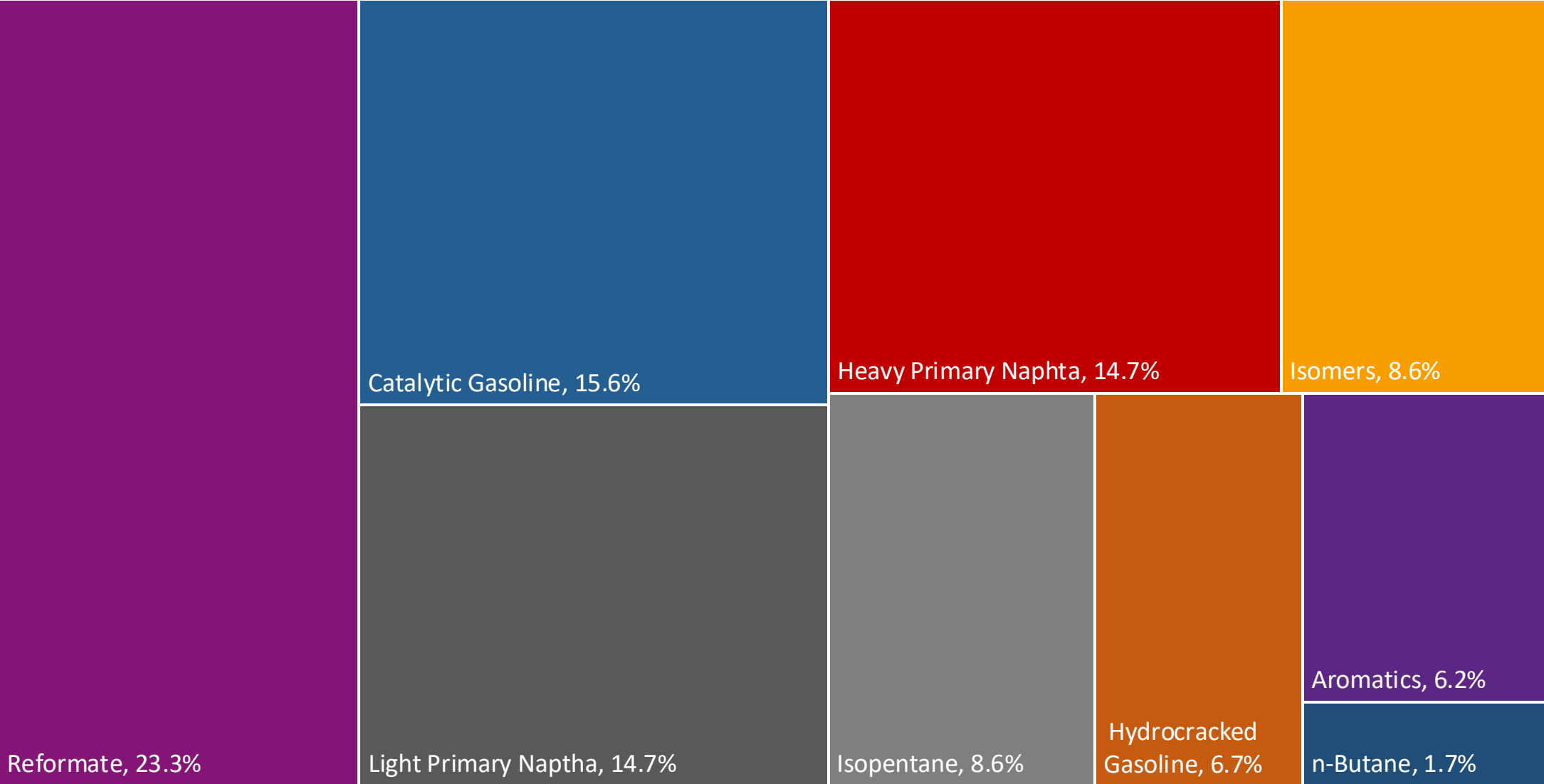
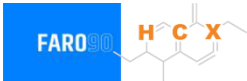
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

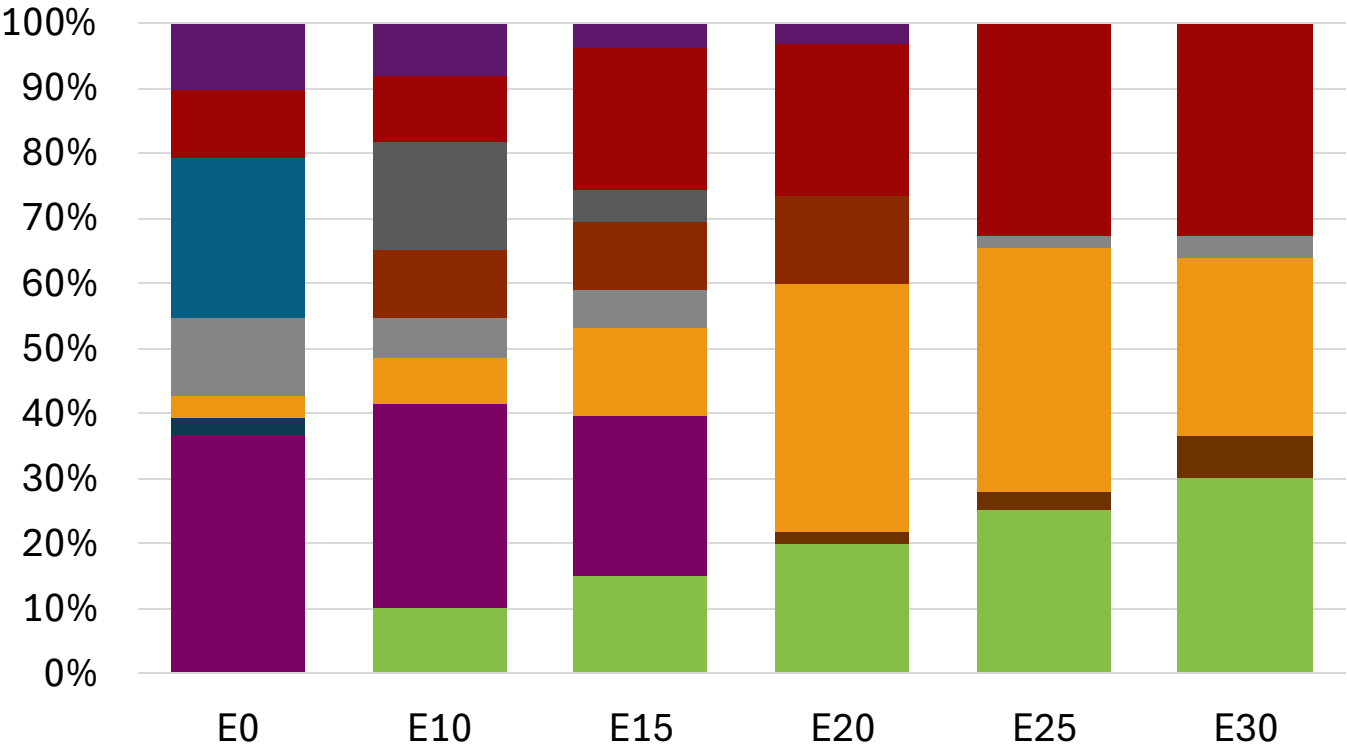
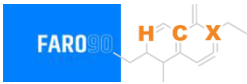


Thailand Available Blending Components



Source: Faro90

Thailand – Gasoline 91 – Constant Octane

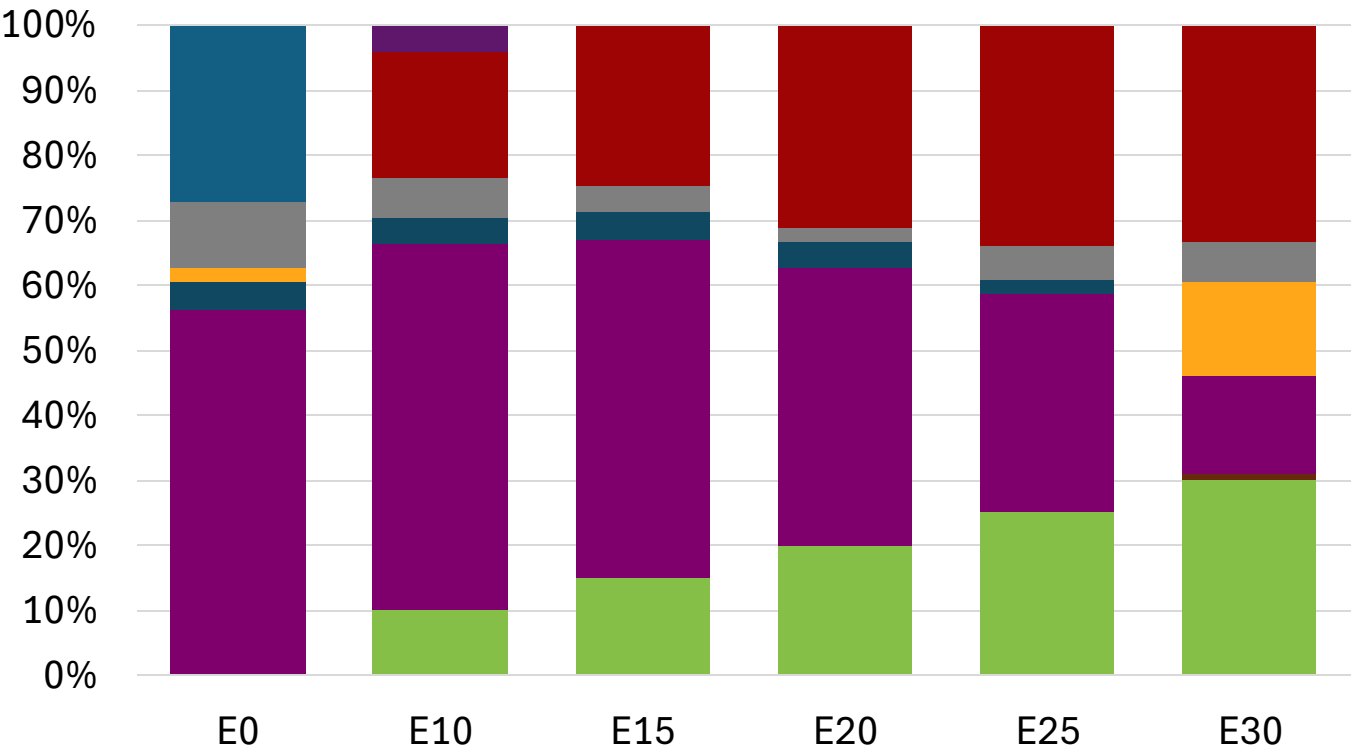
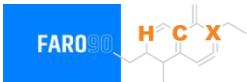


Average CIF 2024 Prices

Octane (RON)	91.0	91.0	91.0	91.4	91.6	93.1
Price (US\$/gal)	\$ 2.256	\$ 2.123	\$ 2.007	\$ 1.947	\$ 1.808	\$ 1.766

Source: Faro90

Thailand - Gasoline 95 - Constant Octane

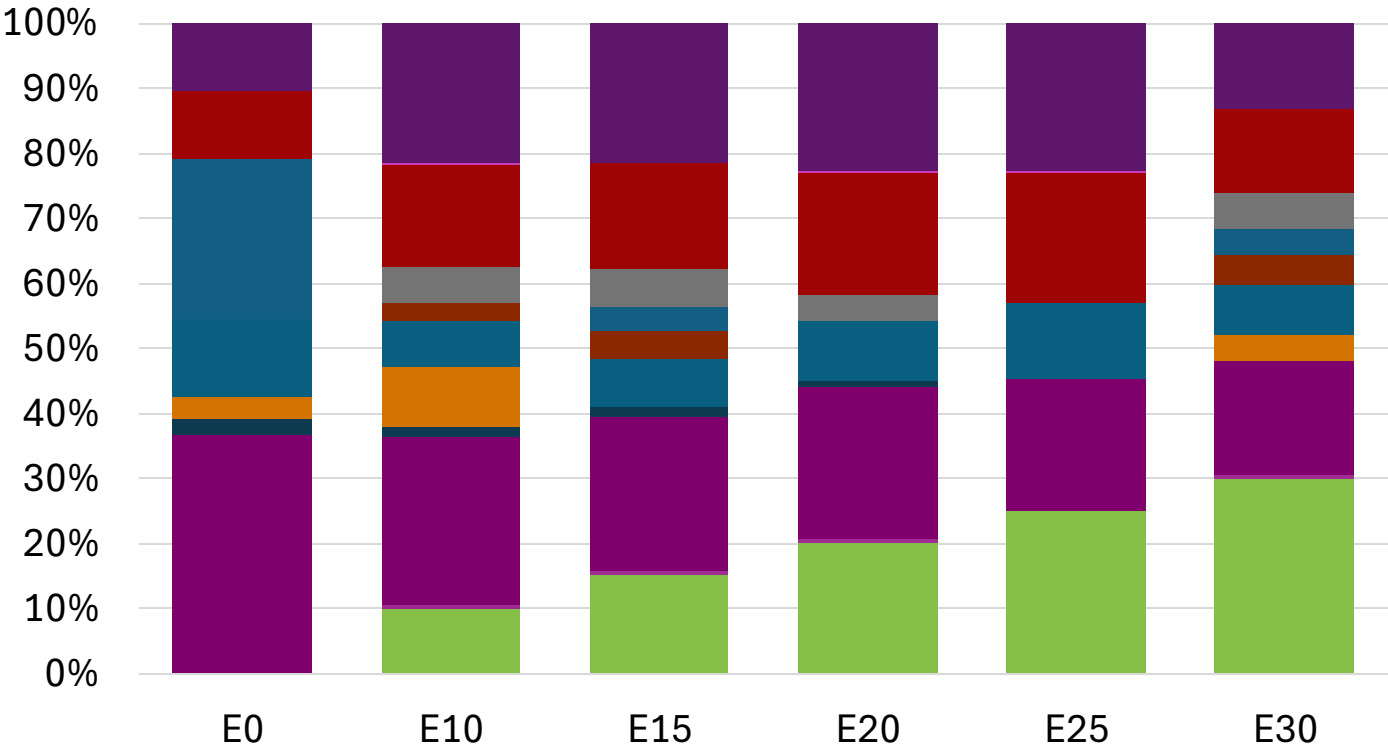
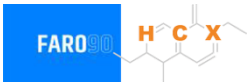


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US\$/gal)	\$ 2.505	\$ 2.276	\$ 2.179	\$ 2.075	\$ 1.968	\$ 1.873

Source: Faro90

Thailand – Gasoline 91 – Increased Octane

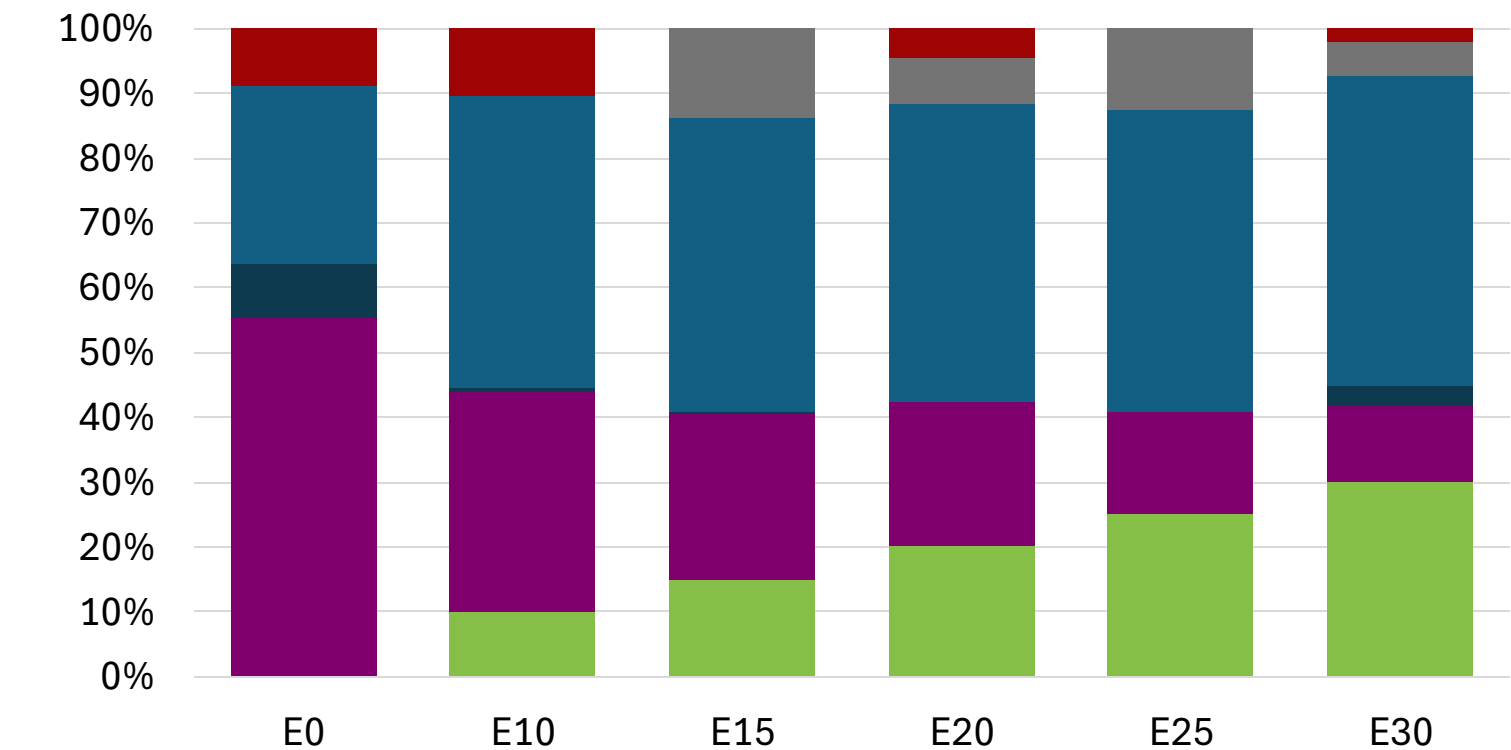
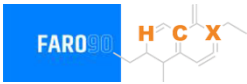


Average CIF 2024
Prices

Octane (RON)	91.0	93.8	95.7	97.5	99.2	100.7
Price (US\$/gal)	\$ 2.256	\$ 2.256	\$ 2.256	\$ 2.256	\$ 2.256	\$ 2.256

Source: Faro90

Thailand – Gasoline 95 – Increased Octane



Average CIF 2024
Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

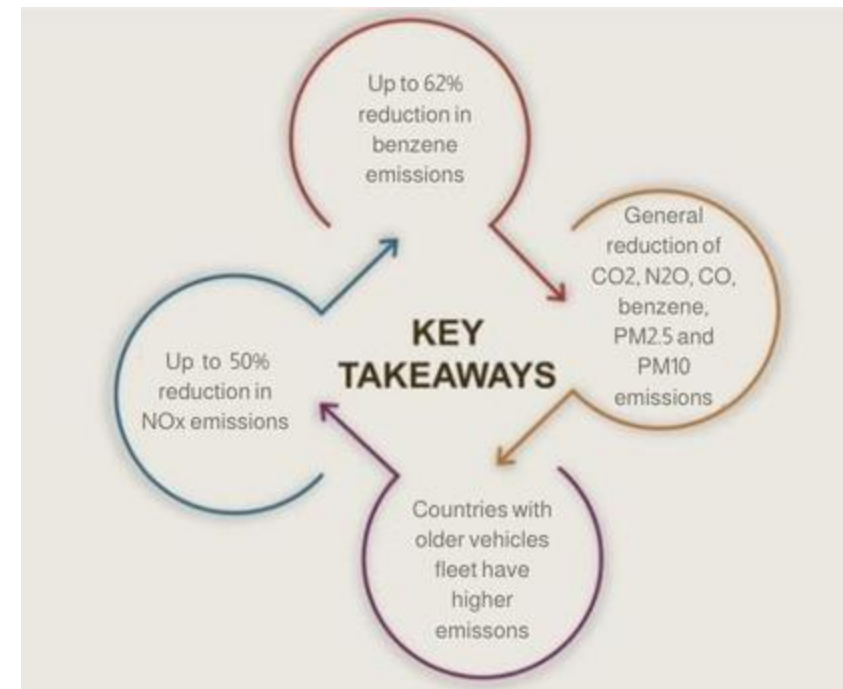
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

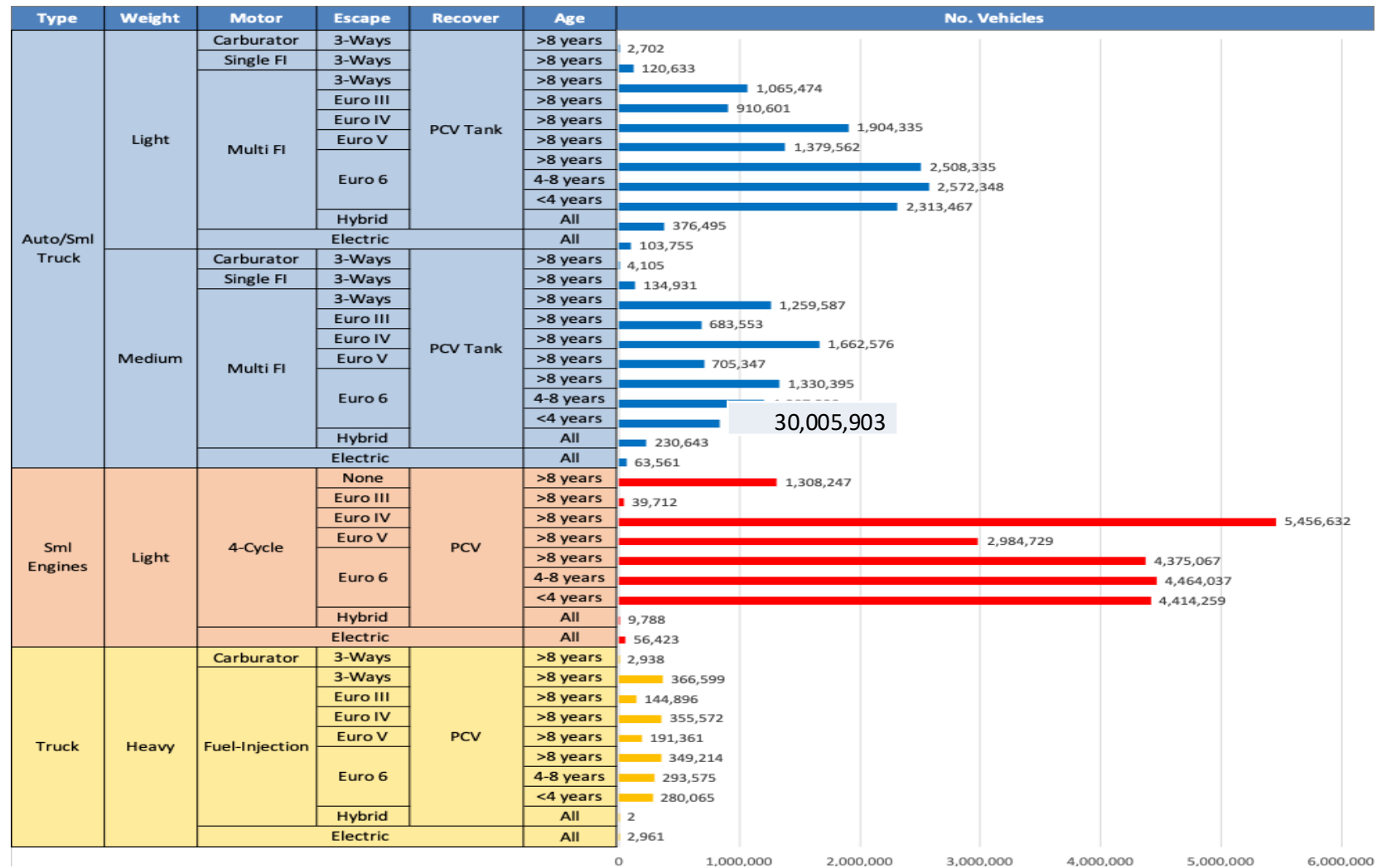
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Thailand

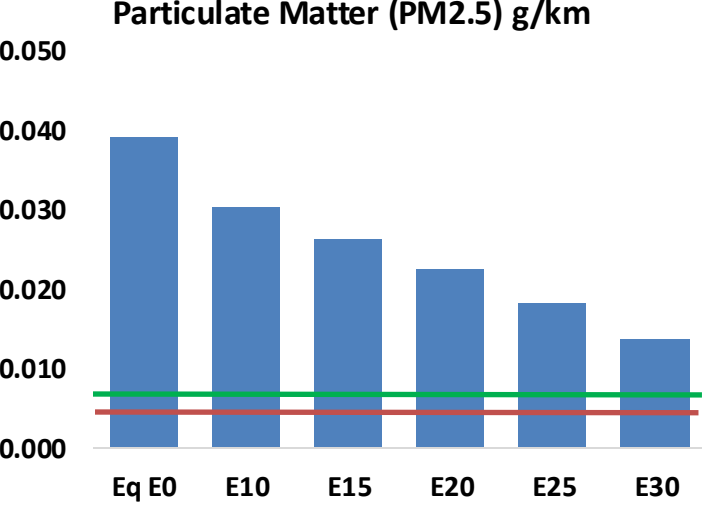
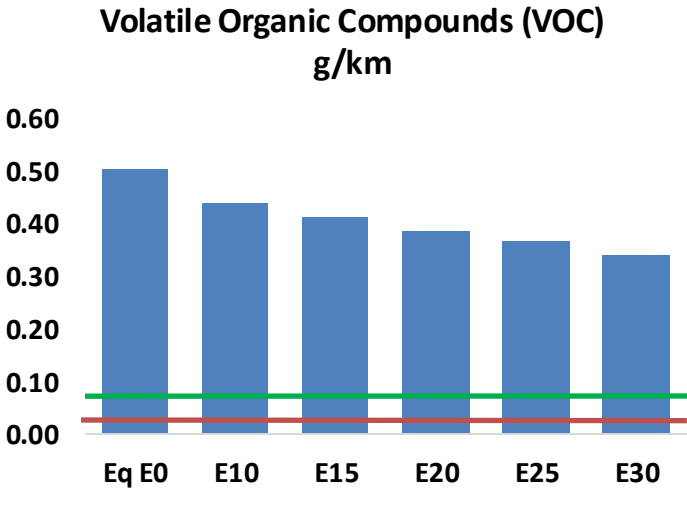
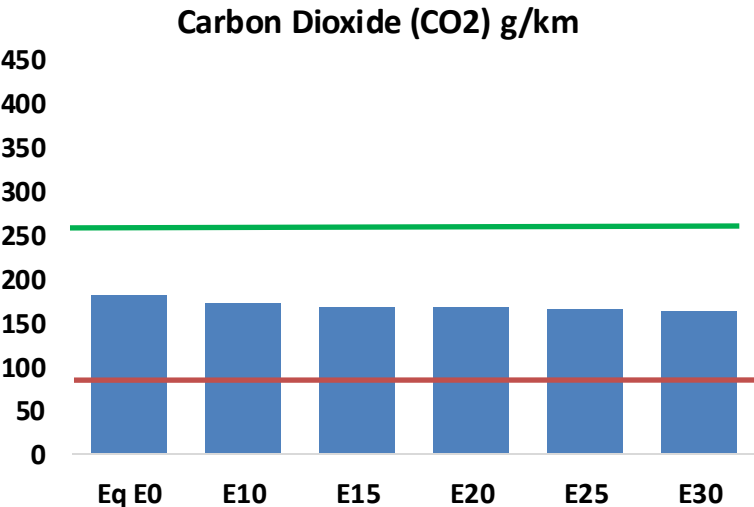
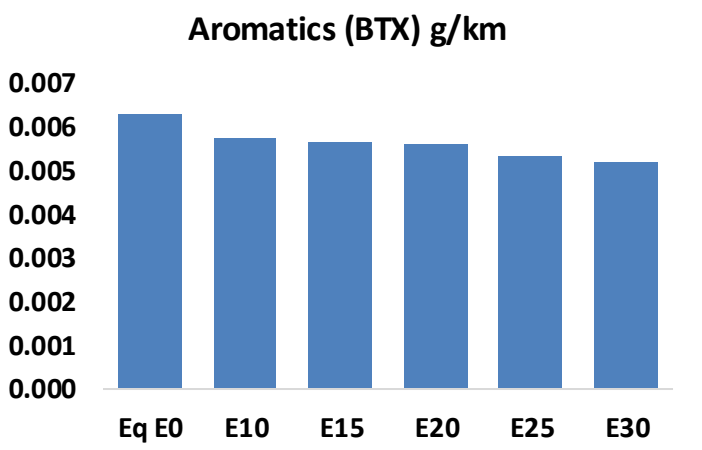
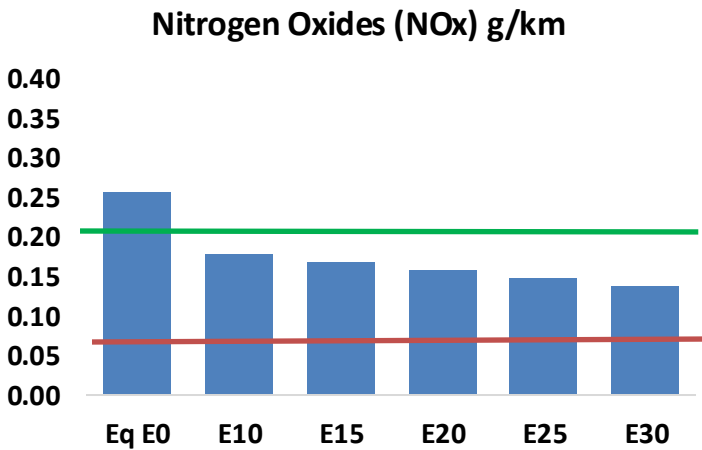
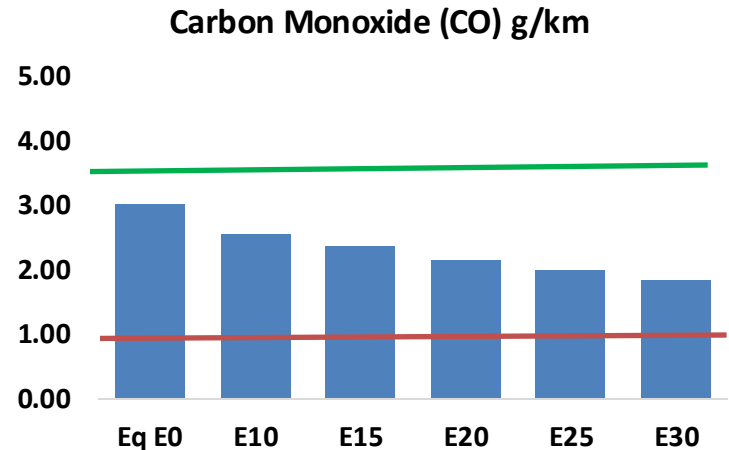
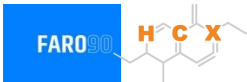


Vehicle Fleet: 46,475,523
Gasoline Vehicle Fleet: 30,005,903

Source: datagov.mot.go.th

Thailand – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Thailand – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,771,402	1,631,957	1,492,513	1,377,607	1,261,681	1,172,451	1,075,545	-16%	-29%
CO2	105,993,888	103,268,331	100,694,194	98,669,711	97,667,917	97,249,062	95,456,482	-5%	-8%
VOC	294,251	275,158	256,065	241,107	226,372	215,137	199,807	-13%	-23%
NOx	150,376	112,782	105,263	99,211	93,564	87,339	80,752	-30%	-38%
SOx	1,202	1,111	1,020	943	870	790	706	-15%	-28%
NH3	40,484	40,083	39,682	39,871	40,319	40,635	40,897	-2%	0%
N2O	1,710	1,701	1,694	1,703	1,738	1,757	1,777	-1%	2%
CH4	70,833	70,833	70,833	71,896	73,454	74,800	76,075	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,944	1,844	1,745	1,673	1,605	1,554	1,482	-10%	-17%
Acetaldehyde	5,608	6,437	9,444	14,383	19,595	22,391	26,457	68%	249%
Formaldehyde	21,725	21,121	24,621	28,911	30,288	33,507	36,476	13%	39%
Benzene	3,690	3,546	3,360	3,306	3,279	3,136	3,035	-9%	-11%
PM2.5	6,957	6,177	5,398	4,687	3,980	3,230	2,448	-22%	-43%
PM10	15,967	14,178	12,388	10,758	9,135	7,413	5,618	-22%	-43%

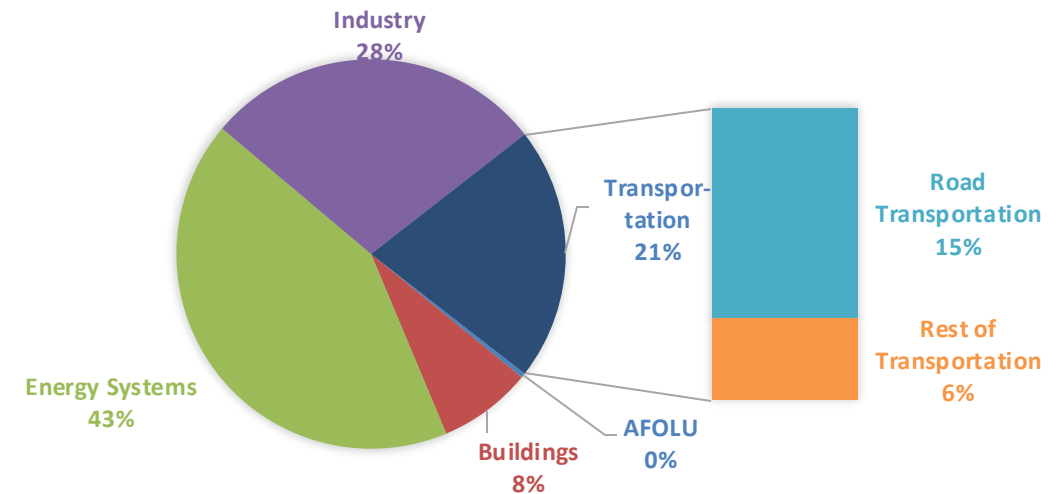
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

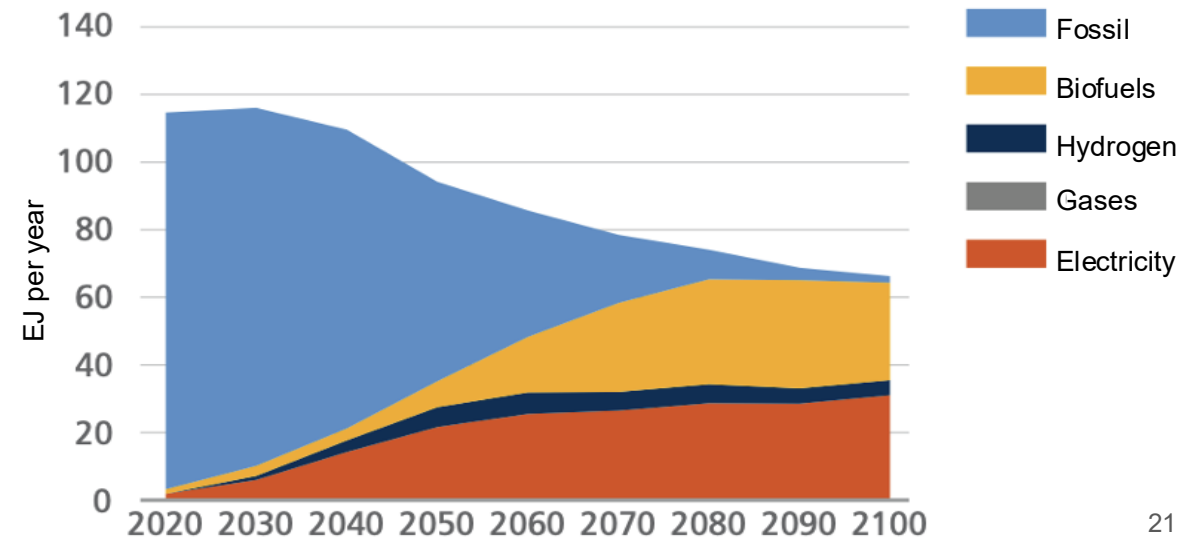
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

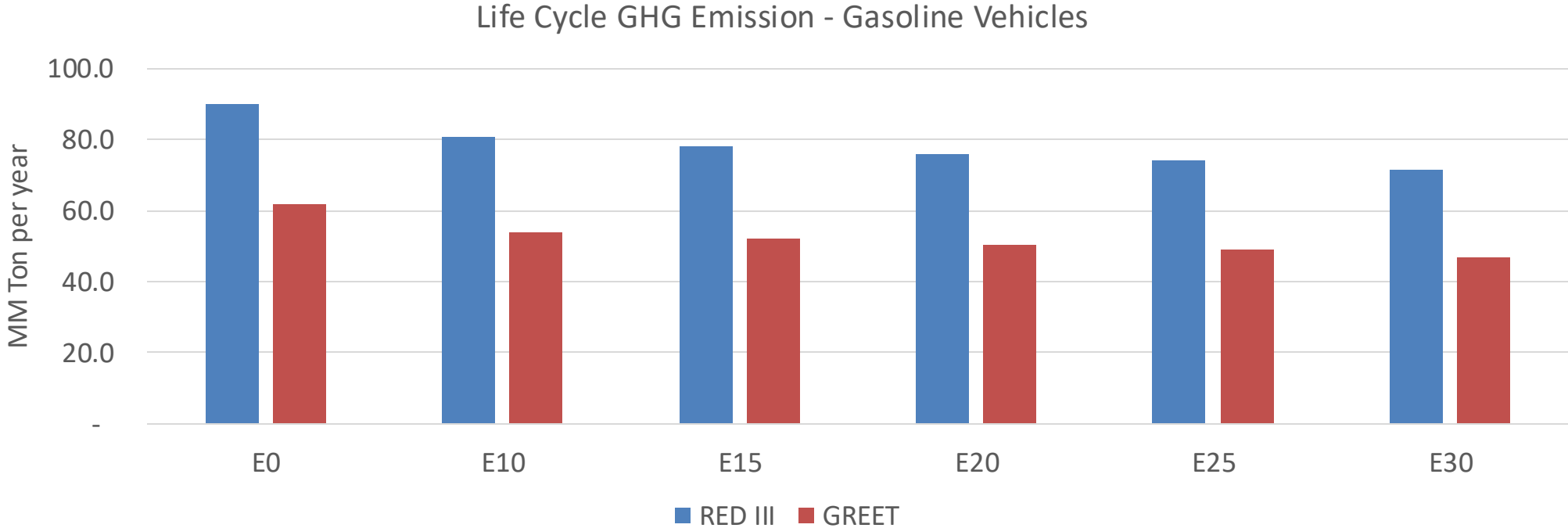
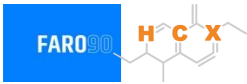
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Thailand – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	356.2
Target @ 2035 (MMT/year)	211.6
Delta	144.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.8%	16.0%	18.5%	20.8%	24.0%
RED II/III	0.0%	10.2%	13.1%	15.5%	17.8%	20.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.5%	6.8%	7.9%	8.9%	10.3%
RED II/III	0.0%	6.4%	8.2%	9.7%	11.1%	12.8%

Source: Greet, RED II/III, climateactiontracker.org, F90